05-17 第一篇

The Mystery of Photosynthesis: How Plants Feed the World

Have you ever wondered how plants can grow without eating food like animals do? The answer lies in a wonderful process called photosynthesis. [1] In fact, it is through this process that plants use sunlight, water, and carbon dioxide from the air to make their own food—sugar. At the same time, they release oxygen into the air, which is what we breathe. That is why plants are often called the "lungs" of our planet. Without photosynthesis, life on Earth as we know it would not exist.

How does it work? [2] The process mainly happens in the leaves, where a green substance called chlorophyll（叶绿素）traps sunlight. Chlorophyll gives plants their green colour. When sunlight hits the leaf, the plant uses its energy to combine water from the roots with carbon dioxide from the air. The result is glucose（葡萄糖）—a kind of sugar that gives the plant energy—and oxygen, which leaves through tiny pores（气孔）on the leaf surface. So leaves are like small factories working day and night.

Why is sunlight so important? [3] Without light, photosynthesis cannot take place. That is why plants grow better in bright places and may turn pale or weak in dark corners. Some plants, like sunflowers, even turn their heads to follow the sun across the sky. In winter, when days are shorter and sunlight is weaker, many plants slow down or lose their leaves. Light is the power source that drives the whole process.

Plants also need water and air. [4] Water travels from the roots up through the stem to the leaves. Carbon dioxide enters through the pores on the underside of leaves. If a plant does not get enough water, its leaves may wilt（枯萎）and photosynthesis slows. If the air is polluted, plants may struggle to get clean carbon dioxide. In other words, clean air and enough water are necessary for healthy photosynthesis.

[5] In short, photosynthesis is one of the most important processes on Earth. It feeds not only plants but also the animals that eat plants, and it provides the oxygen we breathe. By protecting forests and planting more trees, we help keep this amazing process working for all of us. Remember: every green leaf is a tiny power station keeping our world alive.

选项

A. That is because plants make food and oxygen through a special process using light. B. In other words, the leaf is the main place where this "magic" happens. C. That is why light is the key factor in photosynthesis. D. It is true that plants need both water and carbon dioxide to complete the process. E. In conclusion, photosynthesis supports life on our planet in many ways. F. For example, some plants can grow in water instead of soil.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：总领光合作用的核心功能。空前设问"Have you ever wondered how plants can grow without eating food like animals do? The answer lies in a wonderful process called photosynthesis"，空后说明"In fact, it is through this process that plants use sunlight, water, and carbon dioxide...to make their own food—sugar...they release oxygen into the air...plants are often called the 'lungs' of our planet...Without photosynthesis, life on Earth...would not exist"。A选项"That is because plants make food and oxygen through a special process using light"（那是因为植物通过一种利用光的特殊过程制造食物和氧气）中"That is because"解释光合作用的本质，"make food and oxygen...using light"与空后糖、氧气及阳光、水、二氧化碳的利用精准对应，衔接顺畅。
- **[2] B**：
  - **线索**：光合作用发生的主要场所。空前设问"How does it work?"，空后详述"The process mainly happens in the leaves, where a green substance called chlorophyll traps sunlight...Chlorophyll gives plants their green colour...When sunlight hits the leaf...water from the roots...carbon dioxide from the air...result is glucose...oxygen...So leaves are like small factories working day and night"。B选项"In other words, the leaf is the main place where this 'magic' happens"（换言之，叶子是这一"魔法"发生的主要场所）中"In other words"概括空后内容，"leaf...main place...this 'magic'"与空后"mainly happens in the leaves"及"leaves are like small factories"完美呼应，衔接自然。
- **[3] C**：
  - **线索**：阳光对光合作用的关键作用。空前设问"Why is sunlight so important?"，空后说明"Without light, photosynthesis cannot take place...plants grow better in bright places...may turn pale or weak in dark corners...sunflowers...turn their heads to follow the sun...In winter...days are shorter...plants slow down or lose their leaves...Light is the power source that drives the whole process"。C选项"That is why light is the key factor in photosynthesis"（因此阳光是光合作用的关键因素）中"That is why"承接空前设问及空后结论，"light...key factor"与空后"power source that drives the whole process"精准对应，逻辑紧密。
- **[4] D**：

- **线索**：水与二氧化碳的必要性。空前提出"Plants also need water and air"，空后详述"Water travels from the roots up through the stem to the leaves...Carbon dioxide enters through the pores...If a plant does not get enough water, its leaves may wilt and photosynthesis slows...If the air is polluted, plants may struggle to get clean carbon dioxide...clean air and enough water are necessary for healthy photosynthesis"。D选项"It is true that plants need both water and carbon dioxide to complete the process"（的确，植物需要水和二氧化碳来完成这一过程）中"It is true that"加强衔接，"water and carbon dioxide"与空后两大物质及"necessary for healthy photosynthesis"完美呼应，衔接得当。

- **[5] E**：
  - **线索**：全文收束与光合作用意义的总结。空[5]位于结尾段段首，空后总结"photosynthesis is one of the most important processes on Earth...It feeds not only plants but also the animals that eat plants, and it provides the oxygen we breathe...By protecting forests and planting more trees...every green leaf is a tiny power station keeping our world alive"。E选项"In conclusion, photosynthesis supports life on our planet in many ways"（总之，光合作用在许多方面支撑着我们地球上的生命）用"In conclusion"总结全文，"supports life...in many ways"与空后喂养植物与动物、提供氧气、护林植树形成呼应，收束有力。

- **干扰项分析**：
  - **F. For example, some plants can grow in water instead of soil.**：涉及水生植物生长环境，易被误填[2]或[4]。但F讨论的是植物生长介质（水 vs 土），而空[2]聚焦光合作用发生的场所（叶子），空[4]强调光合作用所需的原料（水与二氧化碳）；F属于生长条件差异，与空[2]的场所、空[4]的原料需求论述角度不同。F属"主题相关但论述维度错位"型干扰。